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SAVEETHA INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES
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CSA0203- C PROGRAMMING

Student Result Processing System And Grade Generation File Based Record Processing

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Introduction

- Student Result Processing and Grade Generation System is developed using C programming.
- The project uses File-Based Record Processing to store student records permanently.
- It automates the calculation of total marks, percentage, and grade.
- Student records can be added, viewed, searched, updated, and deleted.
- The system provides a menu-driven interface for easy operation.
- It uses Structures, File Handling, Functions, Loops, and Conditional Statements in C.

Problem Statement

- Student result processing is often done manually, which is time-consuming.
- Manual calculation of total marks, percentage, and grades can lead to errors.
- Maintaining student records on paper is difficult and records may be lost or damaged.
- Searching and updating student information manually is inefficient.
- There is a need for a simple system that stores student records permanently and generates results automatically.

Abstract

- The Student Result Processing and Grade Generation System is a C programming application that simplifies the management of student academic records using file-based record processing.
- The system stores student details permanently, automatically calculates total marks, percentage, and grades, and provides options to add, view, search, update, and delete records.
- It reduces manual effort, improves accuracy, and demonstrates the practical use of C programming concepts such as file handling, structures, functions, and conditional statements in developing a simple student management system.

Existing System

- Student records are maintained manually using paper or registers.
- Result calculation and grade generation are done manually.
- Manual processing is time-consuming and prone to errors.
- Searching and updating student records is difficult.
- Records may be lost or damaged due to improper storage.

Proposed System

- Develop a C-based Student Result Processing System using file handling.
- Store student records permanently in files.
- Automatically calculate total marks, percentage, and grades.
- Provide options to add, view, search, update, and delete student records.
- Reduce manual errors and improve result processing efficiency.

Functionality

- Add new student records.
- View all student records.
- Search student details using Roll Number.
- Update existing student records.
- Delete student records.
- Calculate total marks, percentage, and grade automatically.
- Store and retrieve records using file handling.

Algorithms

- Start the program.
- Display the main menu.
- Select the required operation (Search/Update/Delete).Read or modify student data.
- Calculate total, percentage, and grade.
- Store or retrieve records from the file.
- Display the result.
- Repeat until the user chooses Exit.
- Stop the program.

Conclusion

The project provides a simple and efficient student result management system.

It automates the calculation of total marks, percentage, and grades.

File handling ensures permanent storage of student records.

The system reduces manual effort, minimizes errors, and improves accuracy.

It demonstrates the practical application of C programming and file handling.

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